

Auteur: Rob Willemsen
Studierichting: Informatica
Oefeningenreeks 3A nr. 36.19

$$\begin{aligned}f(x) &= \ln \frac{1-x}{1+x} \\f'(x) &= \frac{1}{1-x} \cdot \frac{(-1)(1+x) - (1-x)1}{(1+x)^2} \\&= \frac{1+x}{1-x} \cdot \frac{-1-x-1+x}{(1+x)^2} \\&= \frac{1+x}{1-x} \cdot \frac{-2}{(1+x)^2} \\&= \frac{-2(1+x)}{(1-x)(1+x)^2} \\&= \frac{-2}{(1-x)(1+x)} \\&= \frac{-2}{1-x^2}\end{aligned}$$

References